20/9/26

Experiment - 1
Basic Networking Commands.

Aim:
To study the Various Network Commands Used in Linux & Windows.

① arp -a ARP is short form of address resolution protocol, It will show the IP address of your computer along with IP address & MAC address of your router.

Output: _ gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3

② hostname: this is the simplest of all TCP/IP commands. It simply displays name of your Computer
output: localhost.localdomain.

③ netstat: (Network Statistics) netstat displays a variety of statistics about a computer's active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routine tables, interface statistics etc.
eg: netstat -r
output:

Destination	Gateway	Genmask	Flags	MSS	Window	irtt	Iface
default	-gateway	0.0.0.0	UG	0	0	0	enp0s3
10.0.2.0	0.0.0.0	255.255.255.0	U	0	0	0	enp0s3

④ nslookup is a tool used to perform DNS lookups in linux. It is used to display DNS details, such as ip address of particular computer, the MX records for a domain or the NS servers of a domain
nslookup can operate in 2 mode: Interactive and non-interactive
eg: nslookup www.google.com
output:
Server: 8.8.8.8
Address: 8.8.8.8#53
Non-authoritative answer:
Name: www.google.com
Address: 142.251.155.119

⑤ ping (packet Internet Groper) command is the best way to test connectivity between two nodes Ping use ICMP (Internet Control Message protocol) to communicate to other devices.

output: pinging google.com [2404:6800:4007:838::200c] with 32 bytes of data.

⑥ pathing: pathing is unique to window and is Basically combination of ping & tracert commands It traces the route to destination address then launches 25 second test of each router
output: Tracing route to google.com
[2404:6800:4013:800::8b]. over maximum of 30 hops

⑦ Route: Route command is used to show the IP routing table. It is primarily used to setup static routes to specific routes

to Specific routes host of network via an interface

O/P Interface List
17: ... 00 00 27 00 00 11 ... VirtualBox Host-only Ethernet Adapter.

IP v4 Route table
Network netmask gateway interface metric
destinator
0.0.0.0 0.0.0.0 172.16.96.1 172.16.102.158

(8) ipconfig/all: This command displays detailed configuration information about your TCP/IP connection including: Router, Router, Gateway, DNS, DHCP; and type of Ethernet adapter in your systems.

OP: enp0s3: flags=4163 <UP Broadcast, Running mtu 1500, inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.55
lo: flags=73<UP, LoopBack, Running> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet 6::1 prefixlen 128 scopeid 0x10 <host>

(9). Some Important Linux Networking Commands.

ip <options> <object> <command>
The ip command is one of the basic commands every administrator will need in daily work, from setting up new systems and assigning IP's to troubleshooting existing systems.

a To show the IP adresses assigned to interface.

a. [root@server ~]

o/p qdisc noqueue qstate unknown group default qlen 1000 link/ loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 124.0.0.118 slope host to valid_lft forever preferred_ft

(b) ip route get
Displays the route taken for IP 8.8.8.
o/p 8.8.8.8 via 172.16.8:1 dev enp2s0 src 172.16.9.200 uid 8479

(c) ip link show - Displays all the network interfaces and shows their status.
o/p 1:10: <LOOPBACK, UP, LOWER-UP> mtu 65536 qdisc qstate unknown mode DEFAULT.

(d) ip route show - Displays system routing table including default gateway, network routes.
output: default via 172.16.81 dev enpsso proto static ~~metric~~ 100

2. ifconfig: This command is a state in many system, tool belt configuring troubleshooting networks. It has been replaced by ip command above
output: flag: 4168<UP, BPODCAST, RUNNING, Multicag mtu 1500. inet 172.16.9.200. netmask 2cc:25c.25.2.0 broadcast 172.16.11.255
inet 6 H02:C677:e632:81b7:7210 prefixlen bH qcopied 0x20<link>

(3) mtr - Matt's traceroute is a program with command line interface that server as a network diagnostic troubleshooting Just like traceroute mtr command mtr provides a lot of statistics about each such as response time, percentage with mtr, you will get more information about route be able to see problematic devices along the way

o/p:

fedora (172.16.9.200) → google.com (142.250.77.110)
2026-07-23T01:44:36-0400

Keys: Help display mode Restart statistics order of fields.

Host	Loss %	Snt	Last	Avg	Best	Wrst	Stdev
gateway	14.7%	200	0.4	0.5	0.2	14.5	2.4

(b) Show numeric Ip address - mtr -g google.com

output: fedora (172.16.9.200) → google.com (142.250.67.46) 2026-07-23T6

Keys: Helps Documentation Restart statistics order of fields

Host	Packetsloss	Snt	Last	Avg	best	Worst
172.16.8.1	0.0%	107	0.3	0.4	4.9	0.5

(c) Show numeric IP with hostname mtr -b google.com

output:

Host	Packets	Snt	Last	Avg	best	Wrst	Dev
gateway	0.0%.	56	8.2	2.7	0.2	12.9	3.3
mail (182.156.9.89)	13.8	84	26.2	21.6	12.5	31.5	4.2

(d) Set the number of pings that you want to send

output:

Host	Packets Loss%	Snt	Avg	Best	Worst
localhost	0.0%	7	0.1	0.0	0.1

(4) tcpdump: is designed for capturing, displaying packets

output: [sudo] password for Linux: root

(5) ping - is a tool verifies IP level connectivity to another TCP/IP Compute by sending Internet Control message $ping google.com

output:

PING google.com (142.250.67.46) 56(44) bytes of data.
64 bytes from maa05s12-in-f14.1e100.net (142.250.67.46): icmp_seq=1 ttl=118 time=8.48ms

Result:

Thus all the networking commands were executed successfully.